機械力学 I 解答用紙

week - 09 減衰自由振動

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m10a02C0202

(1)	$\omega_n = \sqrt{\frac{k}{m}} = 20$ [rad/s]
(2)	$C_c = 2\sqrt{mk} = 200$ [Ns/m]
(3)	$\zeta = \frac{c}{C_c} = 0.10$
(4)	$\omega_d = \sqrt{1-\zeta^2}\,\omega_n \approx 19.9$ [rad/s]
(5)	$f_d = \frac{\omega_d}{2\pi} \approx 3.17$ [Hz]
(6)	$T_d = \frac{1}{f_d} = 0.316$ [s]
(7)	$\delta = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}} \approx 0.631$

フリースペース（質問などどうぞ）

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(1)	$m\ddot{x} + c\dot{x} + kx = 0$
(2)	$\omega_n = \sqrt{\frac{k}{m}}$ [rad/s]
(3)	$\cancel{\zeta = \frac{c}{C_c}}$ $C_c = 2\sqrt{mk}$
(4)	$\zeta = \frac{c}{C_c} = \frac{c}{2\sqrt{mk}}$

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(1)	$m\ddot{x} + c\dot{x} + (k_1+k_2)x = 0$
(2)	$\omega_n = \sqrt{\frac{k_1+k_2}{m}}$
(3)	$C_c = 2\sqrt{m(k_1+k_2)}$
(4)	$\zeta = \frac{c}{C_c} = \frac{c}{2\sqrt{m(k_1+k_2)}}$